



**COMPETENCY STANDARD**

**FOR**

**SUBSTATION MAINTENANCE AND SERVICING**

**(LIGHT ENGINEERING SECTOR)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)**  
**Finance Division, Ministry of Finance**





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The Competency Standards for Substation Maintenance & Servicing is a document for developing curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

This document has been developed to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the skills requirements of the Electrical sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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## INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

**The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.**

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

## OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment, which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

**Units & Elements at a glance:  
Generic Competencies (30 hrs.)**

| <b>Code</b>        | <b>Unit of Competency</b>                              | <b>Elements of Competency</b>                                                                                                                                                                                                     | <b>Duration (Hours)</b> |
|--------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| SICIP-LE-SIM-01-G  | Perform Computations Using Basic Mathematical Formulas | <ol style="list-style-type: none"> <li>1. Identify calculation requirements in the workplace</li> <li>2. Identify and select appropriate mathematical methods</li> <li>3. Use tool/instrument to perform calculations</li> </ol>  | 10                      |
| SICIP-LE-SIM-02-G  | Communicate in English in the Workplace                | <ol style="list-style-type: none"> <li>1. Read, understand and prepare documents in English</li> <li>2. Prepare workplace documents in English</li> <li>3. Perform conversations in English</li> </ol>                            | 10                      |
| SICIP-LE-SIM-03-G  | Operate in a Self-Directed Team                        | <ol style="list-style-type: none"> <li>1. Identify team goals and work processes</li> <li>2. Communicate and cooperate with team members</li> <li>3. Work as a team member</li> <li>4. Solve problems as a team member</li> </ol> | 10                      |
| <b>Total Hours</b> |                                                        |                                                                                                                                                                                                                                   | <b>30</b>               |



## Sector Specific Competencies (30 hrs.)

| Code               | Unit of Competency                                                    | Elements of Competency                                                                                                                                                                                                                                                                                                                      | Duration (Hours) |
|--------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-LE-SIM-01-S  | Apply Occupational Health and Safety (OHS) Practices in the Workplace | <ol style="list-style-type: none"> <li>1. Identify OHS policies and procedures</li> <li>2. Practice personal health and safety procedures</li> <li>3. Report hazards and risks</li> <li>4. Respond to emergencies</li> </ol>                                                                                                                | 10               |
| SICIP-LE-SIM-02-S  | Work with Hand Tools, Power Tools and Electrical Testing Instruments  | <ol style="list-style-type: none"> <li>1. Inspect hand tools, power tools and electrical testing instruments for usability</li> <li>2. Use hand tools and electrical testing instruments properly and safely</li> <li>3. Operate power tools properly and safely</li> <li>4. Clean/maintain hand tools and power tools after use</li> </ol> | 10               |
| SICIP-LE-SIM-03-S  | Apply Green Practices                                                 | <ol style="list-style-type: none"> <li>1. Interpret green concepts</li> <li>2. Minimize resource use in the workplace</li> <li>3. Implement waste management practices</li> </ol>                                                                                                                                                           | 10               |
| <b>Total Hours</b> |                                                                       |                                                                                                                                                                                                                                                                                                                                             | <b>30</b>        |

## Occupation Specific Competencies (300 hrs.)

| Code              | Unit of Competency                             | Elements of Competency                                                                                                                                                                                                                                                                                                                                                                                                                | Duration (hours) |
|-------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-LE-SIM-01-O | Understand basic electrical system             | <ol style="list-style-type: none"> <li>1. Interpret fundamental principles of electricity and electronics</li> <li>2. Identify basic electrical circuit</li> <li>3. Identify cable joints</li> <li>4. Understand different types of electrical loads and load calculation</li> </ol>                                                                                                                                                  | 20               |
| SICIP-LE-SIM-02-O | Understand substation operations               | <ol style="list-style-type: none"> <li>1. Recognize transmission voltage levels</li> <li>2. Comprehend substation equipment</li> <li>3. Interpret voltage transformation system</li> <li>4. Identify high tension (HT) and low tension (LT) cables</li> <li>5. Identify earthing and atmospheric lightning protection systems</li> </ol>                                                                                              | 60               |
| SICIP-LE-SIM-03-O | Assemble and install distribution transformers | <ol style="list-style-type: none"> <li>1. Understand electrical schematics and plan</li> <li>2. Gather necessary tools and equipment</li> <li>3. Assemble the distribution transformer</li> <li>4. Install the distribution transformer</li> <li>5. Connect the transformer to electrical circuits</li> <li>6. Mount and connect switchgear to the transformer</li> <li>7. Test the installation and perform system checks</li> </ol> | 80               |
| SICIP-LE-SIM-04-O | Identify equipment malfunction in substation   | <ol style="list-style-type: none"> <li>1. Perform visual inspection of substation components</li> <li>2. Check transformer for faults</li> <li>3. Check low tension (LT) switchgear</li> </ol>                                                                                                                                                                                                                                        | 80               |

| Code               | Unit of Competency                     | Elements of Competency                                                                                                                                                                                                                                 | Duration (hours) |
|--------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|                    |                                        | 4. Check high tension (HT) switchgear<br>5. Check capacitor bank (power factor improvement) PFI<br>6. Inspect cable connections and earthing                                                                                                           |                  |
| SICIP-LE-SIM-05-O  | Troubleshoot malfunction in substation | 1. Perform initial diagnostic checks<br>2. Test and service substation components<br>3. Check and service control and protection systems<br>4. Perform continuity testing on electrical circuits<br>5. Test transformer and switchgear for performance | 60               |
| <b>Total Hours</b> |                                        |                                                                                                                                                                                                                                                        | <b>300</b>       |

## COMPETENCY STANDARD: SUBSTATION MAINTENANCE AND SERVICING

### A. The Generic Competencies

|                                                                                                                                                                                                                                                                                                                                                                                                  |                                     |                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM COMPUTATIONS USING BASIC MATHEMATICAL FORMULAS</b>                                                                                                                                                                                                                                                                                                      | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-01-G |
| <b>Unit Descriptor:</b><br>This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical concepts in workplace. It specifically includes the tasks of identifying calculation requirements in the workplace, identifying and selecting appropriate mathematical methods, and using appropriate tools and instruments to perform calculations. |                                     |                                        |

#### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| <b>Elements of Competency</b>                           | <b>Performance Criteria</b>                                                                                            |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 1. Identify calculation requirements in the workplace   | 1.1 <b><u>Calculation requirements</u></b> are identified.<br>1.2 <b><u>Workplace information</u></b> are interpreted. |
| 2. Identify and select appropriate mathematical methods | 2.1 <b><u>Appropriate calculation method</u></b> is identified.<br>2.2 Appropriate calculation method is selected.     |
| 3. Use tool/instrument to perform calculations          | 3.1 Appropriate <b><u>tools and instruments</u></b> are identified.<br>3.2 Computation is made.                        |

#### Range of variables:

| <b>Variable</b>                   | <b>Range</b><br>May include but not limited to:                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Calculation requirements       | 1.1 Area<br>1.2 Volume<br>1.3 Measurements: Length/Width/Height/thickness/diameter<br>1.4 Weight and density Calculation                                                                            |
| 2. Workplace information          | 2.1 Drawing<br>2.2 Part drawing<br>2.3 Verbal instructions<br>2.4 Job order and Job sheet                                                                                                           |
| 3. Appropriate calculation method | 3.1 Addition, Subtraction, Division and Multiplication<br>3.2 Conversion of measurement<br>3.3 Percentage and ratio calculation<br>3.4 Geometric Formulas (Square, Rectangle, Circle, Sphere, etc.) |

|                          |                |
|--------------------------|----------------|
| 4. Tools and instruments | 4.1 Calculator |
|                          | 4.2 Computer   |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Numerical concept, identification of calculation requirements<br>1.2 Basic mathematical methods, such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas<br>1.3 Mathematical language, symbols<br>1.4 Measuring units                                                                                                       |
| 2. Underpinning Skills    | 2.1 Computing basic mathematics<br>2.2 Practicing conversion of measurements<br>2.3 Calculating areas, volume, etc. using formulas                                                                                                                                                                                                                                                         |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Tidiness and timeliness<br>3.4 Respect for rights of peers, sub-ordinates and seniors in the workplace<br>3.5 Environmental concerns<br>3.6 Sincerity and honesty to duties<br>3.7 Eagerness to learn<br>3.8 Good relationships with peers, subordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Calculators<br>4.2 Computers<br>4.3 Measuring tape<br>4.4 Stationeries<br>4.5 Consumables                                                                                                                                                                                                                                                 |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                          |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Identified calculation requirements from workplace information.<br>1.2 Selected appropriate method to carry out the calculation requirements.<br>1.3 Completed calculations using appropriate tools/instruments. |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                 |

|                          |                                                                                                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. Context of Assessment | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.</p> |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                         |                                     |                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>COMMUNICATE IN ENGLISH IN THE WORKPLACE</b>                                                                                                                                                                                                                                            | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-02-G |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to communicate in English in the workplace. It specifically includes the tasks of reading, understanding and preparing documents in English, preparing workplace documents in English and performing conversations in English. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                               | Performance Criteria                                                                                                                                                                                                                                         |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Read, understand and prepare documents in English | 1.1 Workplace documents are read and understood.<br>1.2 Visual information is interpreted.<br>1.3 Resume/CV is understood and prepared.                                                                                                                      |
| 2. Prepare workplace documents in English            | 2.1 Simple <b><u>routine workplace documents</u></b> are prepared using key words, phrases and simple sentences.<br>2.2 <b><u>Visual aids</u></b> are prepared and submitted.<br>2.3 Key information is written in the appropriate places in standard forms. |
| 3. Perform conversations in English                  | 3.1 Active listening is demonstrated.<br>3.2 Conversation is performed in English with peers, customers and management as per the workplace standard.                                                                                                        |

### Range of Variables

| Variable                       | Range                                                                                                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | May Include but not limited to:                                                                                                                                                                                                                                 |
| 1. Routine workplace documents | 1.1 Meeting Agenda<br>1.2 Simple reports such as progress and incident reports<br>1.3 Job sheets<br>1.4 Operational manuals<br>1.5 Brochures and promotional materials<br>1.6 Visual and graphic materials<br>1.7 Standards<br>1.8 OSH information<br>1.9 Signs |
| 2. Visual aids                 | 2.1 Maps<br>2.2 Diagrams<br>2.3 Forms<br>2.4 Labels<br>2.5 Graphs<br>2.6 Charts                                                                                                                                                                                 |

**Curricular Content Guide:**

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <ul style="list-style-type: none"><li>1.1 Workplace documents in English</li><li>1.2 Conversation in English</li><li>1.3 Job roles, responsibilities and compliances</li><li>1.4 Understanding of visual aids (e.g., graphs, presentations, or charts) and their preparation</li><li>1.5 Knowledge of workplace communication etiquette (e.g., formal vs. informal conversations)</li><li>1.6 Awareness of workplace standards for professional communication</li></ul>        |
| 2. Underpinning Skills    | <ul style="list-style-type: none"><li>2.1 Reading, understanding and preparing documents in English</li><li>2.2 Preparing workplace documents in English</li><li>2.3 Performing conversations in English</li><li>2.4 Possessing interpersonal skills to maintain polite and professional communication in workplace settings</li></ul>                                                                                                                                         |
| 3. Underpinning Attitudes | <ul style="list-style-type: none"><li>3.1 Commitment to occupational health and safety</li><li>3.2 Promptness in carrying out activities</li><li>3.3 Tidiness and timeliness</li><li>3.4 Respect for rights of peers, sub-ordinates and seniors in the workplace</li><li>3.5 Environmental concerns</li><li>3.6 Sincerity and honesty to duties</li><li>3.7 Eagerness to learn</li><li>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace</li></ul> |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <ul style="list-style-type: none"><li>4.1 Work place procedure</li><li>4.2 Materials relevant to the proposed activity/process</li><li>4.3 Tools, equipment, materials and documentations required</li><li>4.4 Relevant specifications or work instructions</li></ul>                                                                                                                                                         |

**Assessment Evidence Guide:**

|                                   |                                                                                                                                                                                                                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"><li>1.1 Read, understood and prepared documents in English</li><li>1.2 Conversed in English with peers and customers</li><li>1.3 Made reports/documents on workplace activities in English</li></ul> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"><li>2.1 Written test</li><li>2.2 Practical demonstration</li><li>2.3 Oral questioning</li><li>2.4 Portfolio (Optional)</li></ul>                                                                                    |



|                          |                                                                                                                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. Context of Assessment | <p>3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p> |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                               |                                     |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>OPERATE IN A SELF-DIRECTED TEAM</b>                                                                                                                                                                                                                                          | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-03-G |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                         | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify team goals and work processes      | 1.1 Team goals and collaborative decision-making processes are identified.<br>1.2 Roles and responsibilities of team members are identified.<br>1.3 Relationships within team and with other workers are identified.                                                                                                                                                                                                                                                                 |
| 2. Communicate and cooperate with team members | 2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives.<br>2.2 Formal and informal <b><u>forms of communication</u></b> are used effectively to support team achievement.<br>2.3 Diversity in character is respected and valued in team functioning.<br>2.4 Views and opinions of other team members are understood, valued and practiced.<br>2.5 Workplace terminology is used correctly to assist communication. |
| 3. Work as a team member                       | 3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team.<br>3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practice.<br>3.3 Team member's support to other members is appreciated and valued.<br>3.4 Agreed reporting lines are followed using standard operating procedure.                                                                                 |
| 4. Solve problems as a team member             | 4.1 Current and potential problems faced by team are identified.<br>4.2 A solution to the problem is identified.<br>4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.                                                                                                                                                                                                                                                                    |

### Range of Variables

| Variable                  | Range<br>May Include but not limited to:                                                                                                                                                                                                                |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Forms of communication | 1.1 Agenda<br>1.2 Simple reports such as progress and incident reports<br>1.3 Job sheets<br>1.4 Operational manuals<br>1.5 Brochures and promotional materials<br>1.6 Visual and graphic materials<br>1.7 Standards<br>1.8 OSH information<br>1.9 Signs |

### Curricular Content Guide:

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Team goals and collaborative decision-making processes<br>1.2 Roles and responsibilities of team members<br>1.3 Relationships within team and with other workers<br>1.4 Effective interpersonal skills to interact with team members<br>1.5 Effective formal and informal forms of communication<br>1.6 Value of diversity in team functioning<br>1.7 Team's duties, responsibilities, authorities, objectives and task requirements<br>1.8 Support mechanism to other members of team to ensure achievements of goals<br>1.9 Methods of identifying current and potential problems faced by a team<br>1.10 Effective problem-solving methods and evaluation of outcomes |
| 2. Underpinning Skills    | 2.1 Identifying team goals and collaborative decision-making processes<br>2.2 Identifying roles and responsibilities of team members<br>2.3 Using effective interpersonal skills to interact with team members and to contribute to activities and achieving objectives<br>2.4 Understanding and valuing views and opinions of other team members<br>2.5 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice                                                                                                                                                                                                     |

|                           |                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 2.6 Appreciating and valuing team member's support to other members<br>2.7 Identifying current and potential problems faced by the team<br>2.8 Solving problems effectively and evaluating the outcome of the implemented solutions                                                                                                                                                        |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Tidiness and timeliness<br>3.4 Respect for rights of peers, sub-ordinates and seniors in the workplace<br>3.5 Environmental concerns<br>3.6 Sincerity and honesty to duties<br>3.7 Eagerness to learn<br>3.8 Good relationships with peers, subordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Workplace procedure<br>4.3 Materials relevant to the proposed activity<br>4.4 Tools, equipment and facilities required<br>4.5 Relevant specifications or work instructions<br>4.6 Learning manuals                                                                                                 |

#### **Assessment Evidence Guide:**

|                                   |                                                                                                                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Identified team goals and work processes<br>1.2 Communicated and cooperated with team members<br>1.3 Worked as a team member<br>1.4 Solved problems as a team member                        |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                            |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert. |

## B. The Sector Specific Competencies

|                                                                                                                                                                                                                                                                                                                                                            |                                     |                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE</b>                                                                                                                                                                                                                                                 | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-01-S |
| <b>Unit Descriptor:</b><br>This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks and responding to emergencies. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                            | Performance Criteria                                                                                                                                                                                                                                 |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify OHS policies and procedures           | 1.1 <b><u>OHS policies</u></b> and safe operating procedures are read and understood.<br>1.2 Safety signs and symbols are identified and followed.<br>1.3 Emergency response, evacuation procedures and other contingency measures are determined.   |
| 2. Practice personal health and safety procedures | 2.1 OHS policies and procedures are followed and practiced.<br>2.2 <b><u>Personal Protective Equipment (PPE)</u></b> are selected and used.<br>2.3 Personal health, hygiene and safety procedures are practiced.                                     |
| 3. Report hazards and risks                       | 3.1 <b><u>Hazards and risks</u></b> are identified, assessed and controlled.<br>3.2 Incidents arising from hazards and risks are reported to the authority.<br>3.3 Corrective actions are implemented to correct unsafe conditions in the workplace. |
| 4. Respond to emergencies                         | 4.1 Alarms and warning devices are responded.<br>4.2 <b><u>Emergency response plans and procedures</u></b> are implemented.<br>4.3 <b><u>First aid procedure</u></b> is applied during emergency situations.                                         |

### Range of Variables

| Variable        | Range                                                                              |
|-----------------|------------------------------------------------------------------------------------|
|                 | May include but not limited to:                                                    |
| 1. OHS policies | 1.1 International/ Local OHS requirements<br>1.2 Fire Safety Rules and Regulations |

|                                            |                                                                                                                                                                                  |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | 1.3 Industry Guidelines                                                                                                                                                          |
| 2. Personal Protective Equipment (PPE)     | 2.1 Apron<br>2.2 Gloves<br>2.3 Safety shoes<br>2.4 Helmet<br>2.5 Face mask<br>2.6 Goggles and safety glasses<br>2.7 Ear plugs<br>2.8 Sun block<br>2.9 Chemical/Gas masks         |
| 3. Hazards and risks                       | 3.1 Chemical hazards<br>3.2 Biological hazards<br>3.3 Physical Hazards 3.3.1 Machine hazards<br>3.3.2 Materials hazards<br>3.3.3 Tools and Equipment hazards                     |
| 4. Emergency response plans and procedures | 4.1 Firefighting procedures<br>4.2 Earthquake response procedures<br>4.3 Evacuation procedures<br>4.4 Medical and first-aid                                                      |
| 5. First aid procedure                     | 5.1 Washing of open wound<br>5.2 Washing chemically infected area<br>5.3 Applying bandage<br>5.4 Applying CPR (Cardiopulmonary Resuscitation)<br>5.5 Taking appropriate medicine |

#### **Curricular Content Guide:**

|                           |                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 OHS workplace policies and procedures<br>1.2 Work safety procedures<br>1.3 Emergency procedures 1.3.1 Firefighting<br>1.3.2 Earthquake response<br>1.3.3 Explosion response<br>1.3.4 Accident response<br>1.4 Types of hazards (biological, chemical and physical) and their effects<br>1.5 PPE types and uses<br>1.6 Personal hygiene practices<br>1.7 Awareness of OHS |
| 2. Underpinning Skills    | 2.1 Identifying OHS policies and procedures<br>2.2 Following personal work safety practices<br>2.3 Reporting hazards and risks<br>2.4 Responding to emergency procedures<br>2.5 Maintaining physical well-being in the workplace                                                                                                                                             |

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|                           | 2.6 Using firefighting accessories and fire extinguishers<br>2.7 Applying basic first aid procedures                                                                                                                                                                                                                                                                         |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Tools, equipment and facilities appropriate to the process or activities<br>4.3 Materials relevant to the proposed activity<br>4.4 PPEs<br>4.5 Firefighting equipment<br>4.6 Emergency response manual<br>4.7 First aid kits                                                         |

#### **Assessment Evidence Guide:**

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Followed OHS policies and procedures<br>1.2 Selected and used personal protective equipment (PPE)<br>1.3 Practiced personal health and safety procedures<br>1.4 Reported incidents arising from hazards and risks to the authority<br>1.5 Implemented plans and procedures to respond emergency<br>1.6 Applied basic first aid procedure |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                                                                                                                                         |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.                                                                                                                                              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                     |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>WORK WITH HAND TOOLS, POWER TOOLS AND ELECTRICAL TESTING INSTRUMENTS</b>                                                                                                                                                                                                                                                                                                                                                                       | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-02-S |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to work with hand tools, power tools and electrical testing instruments. It specifically includes the tasks of inspecting hand tools, power tools and electrical testing instruments for usability, using hand tools and electrical testing instruments properly and safely, operating power tools properly and safely, and cleaning/maintaining hand tools and power tools after use. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                                                              | Performance Criteria                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Inspect hand tools, power tools and electrical testing instruments for usability | 1.1 Appropriate tools are selected.<br>1.2 Application of tools to job requirement is determined.<br>1.3 Usability of tools are checked and verified.<br>1.4 <b><u>Hand tools and electrical testing instruments/measuring devices</u></b> are prepared.<br>1.5 Sources of power supply for power tools identified.                      |
| 2. Use hand tools and electrical testing instruments properly and safely            | 2.1 Appropriate hand tools, electrical testing instruments for the job are used.<br>2.2 Proper and safe use/operation is applied in the different types of hand tools.<br>2.3 <b><u>Safety precautions</u></b> is observed when using hand tools.<br>2.4 Unsafe or faulty tools are identified and marked for repairing.                 |
| 3. Operate power tools properly and safely                                          | 3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements.<br>3.2 Proper sequence of operation is applied in using power tools to produce results.<br>3.3 Power tools are used safely in accordance with manufacturer's operating specification. |
| 4. Clean/maintain hand tools and power tools after use                              | 4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard.<br>4.2 Scraps or wastages are removed and cleaned.<br>4.3 Condition of tools is checked after use.<br>4.4 Appropriate lubricant is applied after use and prior to storage.<br>4.5 Measuring tools are checked and calibrated.         |



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|  | 4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced. |
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### Range of Variables

| Variable                                                            | Range                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | May include but not limited to:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1. Hand tools and electrical testing instruments/ measuring devices | <p>1.1 Tools:</p> <p>1.1.1 Side cutter pliers</p> <p>1.1.2 Set of screw drivers</p> <p>1.1.3 Set of combination wrenches</p> <p>1.1.4 Adjustable wrench</p> <p>1.1.5 Wire stripper</p> <p>1.1.6 Electrician's knife</p> <p>1.1.7 Measuring tape</p> <p>1.1.8 Electrical Hand Drill</p> <p>1.2 Electrical testing Instruments:</p> <p>1.2.1 Multi-meter/AVO meter</p> <p>1.2.2 Clamp meter</p> <p>1.2.3 Volt meter</p> <p>1.2.4 Ammeter</p> <p>1.2.5 Power meter</p> <p>1.2.6 Megger tester</p> <p>1.2.7 Soldering iron with stand</p> <p>1.2.8 Solder sucker</p> <p>1.2.9 Neon tester</p> <p>1.3 Mechanical Tools:</p> <p>1.3.1 Cutting pliers</p> <p>1.3.2 Nose pliers</p> <p>1.3.3 Combination pliers</p> <p>1.3.4 Measuring tape</p> <p>1.3.5 Tweezers</p> <p>1.3.6 Scissors</p> <p>1.3.7 Wire striper</p> <p>1.3.8 Power Saw</p> <p>1.3.9 Drill Machine</p> |
| 2. Safety precautions                                               | <p>2.1 Use of appropriate PPEs</p> <p>2.2 Proper hand, feet and eye coordination</p> <p>2.3 Working environment</p> <p>2.4 Safe operating condition for hand tools and power tools</p> <p>2.5 Awareness of OHS requirements</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <ul style="list-style-type: none"> <li>1.1 Functions of different types of tools and their uses</li> <li>1.2 Proper use and techniques of different types of hand tools</li> <li>1.3 Functions of different types of electrical testing instruments</li> <li>1.4 Types of Power tools, use and safe handling method</li> <li>1.5 Technical application of tools</li> <li>1.6 Procedures in the use of hand tools and power tools</li> <li>1.7 Reporting and documentation</li> <li>1.8 Safety precautions</li> <li>1.9 Quality procedures</li> <li>1.10 Management of scraps or wastages</li> <li>1.11 Storage procedures</li> </ul>                                                                                                  |
| 2. Underpinning Skills    | <ul style="list-style-type: none"> <li>2.1 Using appropriate hand tools for the job</li> <li>2.2 Observing safety precautions when using hand tools</li> <li>2.3 Using electrical testing instruments or devices</li> <li>2.4 Using power tools correctly and safely in accordance with manufacturer's operating specification</li> <li>2.5 Checking condition of tools after use</li> <li>2.6 Applying appropriate lubricant on hand tools and power tools after use and prior to storage</li> <li>2.7 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories</li> <li>2.8 Storing Tools, electrical testing instruments/devices and power tools safely in appropriate location</li> </ul> |
| 3. Underpinning Attitudes | <ul style="list-style-type: none"> <li>3.1 Commitment to occupational health and safety</li> <li>3.2 Promptness in carrying out activities</li> <li>3.3 Sincerity and honesty to duties</li> <li>3.4 Environmental concerns</li> <li>3.5 Eagerness to learn</li> <li>3.6 Tidiness and timeliness</li> <li>3.7 Respect for rights of peers, sub-ordinates and seniors in the workplace</li> <li>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace</li> </ul>                                                                                                                                                                                                                                               |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <ul style="list-style-type: none"> <li>4.1 Workplace (simulated or actual)</li> <li>4.2 Different types of Light Engineering hand tools and power tools</li> <li>4.3 Operation and maintenance manuals</li> <li>4.4 Tools, equipment and facilities appropriate to the process or activities</li> <li>4.5 Materials relevant to the proposed activity</li> <li>4.6 Work books</li> </ul>                                                                                                                                                                                                                                                                                             |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Used appropriate hand tools for the job<br>1.2 Observed safety precautions when using hand tools<br>1.3 Used power tools safely in accordance with manufacturer's operating specification<br>1.4 Cleaned and maintained hand tools and power tools after use and prior to storage<br>1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories<br>1.6 Stored tools and power tools safely in appropriate location |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written Test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                                                                                                                                                                                                                                                          |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.                                                                                                                                                                                                                                                     |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY GREEN PRACTICES</b>                                                                                                                                                                                                                  | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-03-S |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                    | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret green concepts               | 1.1 <b><u>Principles of green practices</u></b> are explained.<br>1.2 <b><u>Sources of environmental impacts</u></b> during light engineering activities are identified.<br>1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained.<br>1.4 <b><u>Ways of mitigating environmental impacts</u></b> in light engineering sector are explained. |
| 2. Minimize resource use in the workplace | 2.1 Water, energy and raw material consumption are documented.<br>2.2 <b><u>Recyclable and non-recyclable items</u></b> are identified.<br>2.3 Procedures to reduce resource consumption are implemented.<br>2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.                                                                                                                       |
| 3. Implement waste management practices   | 3.1 <b><u>Different types of wastes</u></b> are identified.<br>3.2 Hazardous waste is disposed of according to environmental regulations.<br>3.3 Green habits to reduce waste in personal and professional life are practiced.                                                                                                                                                                                             |

### Range of Variables

| Variable                         | Range                                                                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | May include but not limited to:                                                                                                                                 |
| 1. Principles of green practices | 1.1 Reducing energy consumption<br>1.2 6R for waste management<br>1.2.1 Refuse<br>1.2.2 Reduce<br>1.2.3 Reuse<br>1.2.4 Recycle<br>1.2.5 Recover<br>1.2.6 Repair |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | 1.3 use of sustainable materials with low environmental impact<br>1.4 Recycling materials<br>1.5 Sustainable transportation                                                                                                                                                                                                                                                                                                                 |
| 2. Sources of environmental impacts         | 2.1 Air pollution<br>2.1.1 Dust generation<br>2.1.2 Emission from machinery and vehicles<br>2.2 Water pollution<br>2.2.1 Debris and sediments<br>2.2.2 Chemical's leakage<br>2.3 Noise pollution<br>2.4 Waste generation<br>2.4.1 waste<br>2.4.2 Non-recyclable and hazardous materials<br>2.5 Resource Depletion<br>2.5.1 Excessive use of raw materials<br>2.5.2 Non-renewable material uses, like use of fossil fuel, steel cement, etc. |
| 3. Ways of mitigating environmental impacts | 3.1 Utilizing Energy-Efficient Equipment<br>3.2 Adopting Renewable Energy Sources<br>3.3 Implementing Site Protection Measures<br>3.4 Using Reusable Materials<br>3.5 Choosing Sustainable Materials<br>3.6 Using Noise-Reducing Equipment and scheduling work appropriately<br>3.7 Optimizing Logistics and Delivery                                                                                                                       |
| 4. Recyclable and non-recyclable items      | 4.1 Recyclable items<br>4.1.1 Metal<br>4.1.2 Wood<br>4.1.3 Brick<br>4.1.4 Concrete<br>4.1.5 Plastic<br>4.2 Non-recyclable items<br>4.2.1 Paints and coatings<br>4.2.2 Contaminated materials like lead paint, asbestos<br>4.2.3 Treated wood                                                                                                                                                                                                |
| 5. Different types of wastes                | 5.1 Wood waste<br>5.2 Metal waste<br>5.3 Plastic waste<br>5.4 Electrical system waste<br>5.5 Gypsum board waste<br>5.6 Packaging waste<br>5.7 Organic waste<br>5.8 Chemical West<br>5.9 Fuel/Oil west                                                                                                                                                                                                                                       |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <ul style="list-style-type: none"> <li>1.1 Principles of green practices in Light Engineering sector</li> <li>1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings</li> <li>1.3 Sources of environmental impacts in Light Engineering</li> <li>1.4 Methods to minimize resource consumption (water, energy, raw materials)</li> <li>1.5 Waste management practices</li> <li>1.6 Sustainable alternatives to fossil-based energy resources</li> <li>1.7 Personal and workplace habits for reducing environmental impact</li> </ul>                                                              |
| 2. Underpinning Skills    | <ul style="list-style-type: none"> <li>2.1 Identifying environmental impacts and sources of impacts in Light Engineering activities</li> <li>2.2 Applying methods to minimize resource use in the workplace</li> <li>2.3 Implementing waste management practices</li> <li>2.4 Following procedures to safely disposal of hazardous materials</li> <li>2.5 Documenting water, energy, and material consumption in the workplace</li> <li>2.6 Performing green practices in personal and professional activities</li> <li>2.7 Identifying recyclable and non-recyclable items</li> <li>2.8 Identifying different types of wastes</li> </ul> |
| 3. Underpinning Attitudes | <ul style="list-style-type: none"> <li>3.1 Commitment to occupational health and safety</li> <li>3.2 Promptness in carrying out activities</li> <li>3.3 Sincerity and honesty to duties</li> <li>3.4 Environmental concerns</li> <li>3.5 Eagerness to learn</li> <li>3.6 Tidiness and timeliness</li> <li>3.7 Respect for rights of peers, sub-ordinates and seniors in the workplace</li> <li>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace</li> </ul>                                                                                                                                                   |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <ul style="list-style-type: none"> <li>4.1 Workplace (simulated or actual)</li> <li>4.2 Different types of Light Engineering hand tools and power tools</li> <li>4.3 Operation and maintenance manuals</li> <li>4.4 Waste segregation bins</li> <li>4.5 Tools, equipment and facilities appropriate to the process or activities</li> <li>4.6 Materials relevant to the proposed activity</li> </ul>                                                                                                                                                                                     |

## Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Explained the principles of green practices in Light Engineering sector<br>1.2 Identified sources of environmental impacts<br>1.3 Implemented procedures to minimize resource use (water, energy, raw materials)<br>1.4 Disposed of hazardous waste in compliance with safety and environmental standards<br>1.5 Applied methods to minimize resource use in the workplace<br>1.6 Practiced green habits to reduce waste in both personal and professional life |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                                                                                                                                                                                                                                                                |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.                                                                                                                                                                                                                                                           |

### C. Occupation-specific Competencies

|                                                                                                                                                                                                                                                                                                                                                                                        |                                     |                                        |
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| <b>Unit of Competency:</b><br><b>UNDERSTAND BASIC ELECTRICAL SYSTEM</b>                                                                                                                                                                                                                                                                                                                | <b>Nominal Duration:</b><br>20 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-01-O |
| <b>Unit Descriptor:</b><br>This unit of competency requires knowledge, skills and attitude to understand basic electrical system. It specifically includes the tasks of interpreting fundamental principles of electricity and electronics, identifying basic electrical circuit, identifying cable joints and understanding different types of electrical loads and load calculation. |                                     |                                        |

#### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                                                 | Performance Criteria                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret fundamental principles of electricity and electronics     | 1.1 <b><u>Fundamental principles</u></b> of electricity and electronics are described.<br>1.2 Application of Ohm's law is explained.<br>1.3 <b><u>Electrical/Electronic properties/parameters</u></b> are measured using appropriate measuring tool/instrument or comprehended |
| 2. Identify basic electrical circuit                                   | 2.1 <b><u>Basic elements</u></b> for preparing electrical circuits are identified and collected<br>2.2 <b><u>Electrical/electronic circuit diagram</u></b> is understood and analyzed.<br>2.3 <b><u>Basic electrical circuit</u></b> wirings are described.                    |
| 3. Identify cable joints                                               | 3.1 Different types of <b><u>cable joints</u></b> used in industry are explained.<br>3.2 Characteristics and sizes of the cables are interpreted.                                                                                                                              |
| 4. Understand different types of electrical loads and load calculation | 4.1 <b><u>Resistive loads</u></b> are identified and explained.<br>4.2 <b><u>Inductive loads</u></b> are identified and explained.<br>4.3 <b><u>Capacitive loads</u></b> are identified and explained.                                                                         |

#### Range of Variables

| Variable                  | Range (Includes but not limited to:)                                                                                                                                                                                                           |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Fundamental principles | 1.1 Ohm's Law<br>1.2 Kirchhoff's law<br>1.3 Principles and theory of AC/DC circuits<br>1.4 Series and parallel circuits<br>1.5 Law of conductivity<br>1.6 Law of resistivity<br>1.7 Semiconductor<br>1.8 Half wave and full wave rectification |



|                                                |                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2. Electrical/electronic properties/parameters | 2.1 Voltage<br>2.2 Resistance<br>2.3 Current<br>2.4 Impedance<br>2.5 Capacitance<br>2.6 Resistance<br>2.7 Inductance<br>2.8 Reactance                                                                                                                                                                                |
| 3. Basic elements                              | 3.1 Switch<br>3.2 Socket<br>3.3 Cable<br>3.4 Resistive load<br>3.5 Inductive load                                                                                                                                                                                                                                    |
| 4. Electrical/electronic circuit diagram       | 4.1 Series circuit<br>4.2 Parallel circuit<br>4.3 Series-parallel circuit<br>4.4 Lamp circuits                                                                                                                                                                                                                       |
| 5. Basic electrical circuit                    | 5.1 Series circuits<br>5.2 Parallel circuits<br>5.3 Series-parallel circuits<br>5.4 One Lamp controlled from one point<br>5.5 One Lamp controlled from two points<br>5.6 Two Lamps controlled from two points<br>5.7 Two Lamps controlled from three points<br>5.8 Tube Light Connection<br>5.9 Power supply circuit |
| 6. Cable joints                                | 6.1 Straight Through Joint<br>6.2 Branch Joint<br>6.3 Tee Joint<br>6.4 Tap Joint<br>6.5 Soldered Joint                                                                                                                                                                                                               |
| 7. Resistive loads                             | 7.1 Electric bulb<br>7.2 Heater coil<br>7.3 Electric Iron<br>7.4 Rice Cooker                                                                                                                                                                                                                                         |
| 8. Inductive loads                             | 8.1 Induction motor<br>8.2 Electric fan<br>8.3 Magnetic Conductor<br>8.4 Electric choke coil/Ballast                                                                                                                                                                                                                 |
| 9. Capacitive loads                            | 9.1 Capacitor load bank<br>9.2 PFI plant                                                                                                                                                                                                                                                                             |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <ul style="list-style-type: none"><li>1.1 Fundamental principles of electricity and electronics</li><li>1.2 Application of Ohm's law</li><li>1.3 Electrical/electronic circuit diagram</li><li>1.4 Basic electrical circuit wirings</li><li>1.5 Different types of cable joints used in industry</li><li>1.6 Characteristics and sizes of the cables</li><li>1.7 Resistive loads, inductive loads and capacitive loads</li></ul>                                |
| 2. Underpinning Skills    | <ul style="list-style-type: none"><li>2.1 Measuring electrical/electronic properties/parameters using appropriate measuring tool/instrument</li><li>2.2 Identifying and collecting basic elements for preparing electrical circuits</li><li>2.3 Identifying resistive loads, inductive loads and capacitive loads</li></ul>                                                                                                                                     |
| 3. Underpinning Attitudes | <ul style="list-style-type: none"><li>3.1 Commitment to occupational health and safety</li><li>3.2 Promptness in carrying out activities</li><li>3.3 Sincerity and honesty to duties</li><li>3.4 Environmental concerns</li><li>3.5 Eagerness to learn</li><li>3.6 Tidiness and timeliness</li><li>3.7 Respect for rights of peers and seniors in the workplace</li><li>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace</li></ul> |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <ul style="list-style-type: none"><li>4.1 Workplace (simulated or actual)</li><li>4.2 Tools, equipment and facilities appropriate to the process or activities</li><li>4.3 Various kinds of electrical and electronic components, tools, equipment and instruments</li><li>4.4 Different types and sizes of cables</li></ul>                                                                                   |

## Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"><li>1.1 Described fundamental principles of electricity and electronics</li><li>1.2 Analyzed electrical/electronic circuit diagram</li><li>1.3 Interpreted characteristics and sizes of cables and different types of cable joints</li><li>1.4 Measured electrical/electronic properties/parameters using appropriate measuring tool/instrument</li><li>1.5 Identified and explained resistive loads, inductive loads and capacitive loads</li></ul> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"><li>2.1 Written Test</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                          |                                                                                                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                                                               |
| 3. Context of Assessment | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

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| <b>Unit of Competency:</b><br><b>UNDERSTAND SUBSTATION OPERATIONS</b>                                                                                                                                                                                                                                                                                                                                                          | <b>Nominal Duration:</b><br>60 Hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-02-O |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to understand substation operations. It specifically includes the tasks of recognizing transmission voltage levels, comprehending substation equipment, interpreting voltage transformation system, identifying high tension (HT) and low tension (LT) cables, and identifying earthing and atmospheric lightning protection systems. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                        | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Recognize transmission voltage levels      | 1.1 <b><u>Transmission voltage levels</u></b> are identified and understood.<br>1.2 <b><u>Range of high-voltage transmission levels</u></b> are recognized.<br>1.3 <b><u>Range of medium-voltage transmission levels</u></b> are identified.<br>1.4 Step-up and step-down transformers are recognized.<br>1.5 <b><u>Transformer capacities</u></b> are identified to ensure voltage levels.                                                             |
| 2. Comprehend substation equipment            | 2.1 <b><u>Major equipment</u></b> used in substation are identified and understood.<br>2.2 <b><u>Functions of transformer</u></b> are explained.<br>2.3 Operation of busbars is explained to various sections of the substation.<br>2.4 Surge arresters and lightning protection systems are understood.<br>2.5 Control panels and <b><u>protection relays</u></b> are identified.<br>2.6 Importance of grounding systems in substations is understood. |
| 3. Interpret voltage transformation system    | 3.1 Relationship between primary and secondary voltage is explained.<br>3.2 Structure of transformer windings (both primary and secondary) is understood.<br>3.3 Concept of turns ratio is interpreted.<br>3.4 <b><u>Efficiency losses</u></b> are recognized in the voltage transformation process.                                                                                                                                                    |
| 4. Identify high tension (HT) and low tension | 4.1 High Tension (HT) cables are understood and described.<br>4.2 Low tension (LT) cables are understood and described.                                                                                                                                                                                                                                                                                                                                 |

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| (LT) cables                                                       | 4.3 Specific voltage ratings for HT cables are recognized.<br>4.4 <b>HT cable types</b> are identified, and their suitability for specific applications.<br>4.5 <b>LT cable types</b> are identified, and their suitability for specific applications.       |
| 5. Identify earthing and atmospheric lightning protection systems | 5.1 Earthing systems are identified.<br>5.2 <b>Components of earthing systems</b> are identified and understood.<br>5.3 Atmospheric lightning protection systems (surge protection device) are identified and explained.<br>5.4 Lightning rod is identified. |

### Range of Variables

| Variable                                       | Range (Includes but not limited to:)                                                                                                                                                                                                                    |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Transmission voltage levels                 | 1.1 High to medium voltage transmission levels<br>1.2 Medium to high voltage transmission levels                                                                                                                                                        |
| 2. Range of high-voltage transmission levels   | 2.1 Range 11kV to 400kV<br>2.1.1 11kV<br>2.1.2 33kV<br>2.1.3 132kV<br>2.1.4 230kV<br>2.1.5 400kV                                                                                                                                                        |
| 3. Range of medium-voltage transmission levels | 3.1 Range 400kV to 11kV<br>3.1.1 400kV<br>3.1.2 230kV<br>3.1.3 132kV<br>3.1.4 33kv<br>3.1.5 11kV                                                                                                                                                        |
| 4. Transformer capacities                      | 4.1 KVA<br>4.2 MVA                                                                                                                                                                                                                                      |
| 5. Major equipment                             | 5.1 Transformers<br>5.2 Circuit breakers<br>5.2.1 Vacuum Circuit Breaker (VCB)<br>5.2.2 Load Break Switch (LBS)<br>5.2.3 Air Circuit Breaker (ACB)<br>5.3 High tension switch gear<br>5.4 Low tension switch gear<br>5.5 Power Factor Improvement (PFI) |
| 6. Functions of transformer                    | 6.1 Step-up transformers<br>6.2 Step-down transformers                                                                                                                                                                                                  |
| 7. Protection relays                           | 7.1 Over current protection<br>7.2 Over voltage protection<br>7.3 Under voltage protection<br>7.4 Phase sequence protection                                                                                                                             |

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| 8. Efficiency losses              | 8.1 Core loss<br>8.2 Copper loss                                                    |
| 9. HT cable types                 | 9.1 Overhead cables (35 rm to 300 rm)<br>9.2 Underground cables (35 rm to 300 rm)   |
| 10.LT cable types                 | 10.1 Overhead cables (10 rm to 630 rm)<br>10.2 Underground cables (10 rm to 630 rm) |
| 11.Components of earthing systems | 11.1 Earth electrodes<br>11.2 Conductors<br>11.3 Earth pits                         |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Major equipment used in substation<br>1.2 Functions of transformer<br>1.3 Atmospheric lightning protection systems, surge protection device (surge arrester)<br>1.4 Relationship between primary and secondary voltage<br>1.5 Structure of transformer windings (both primary and secondary), and concept of turn ratio                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2. Underpinning Skills    | 2.1 Identifying transmission voltage levels, range of high-voltage transmission levels and range of medium-voltage transmission levels<br>2.2 Recognizing step-up and step-down transformers<br>2.3 Identifying transformer capacities to ensure voltage levels<br>2.4 Identifying major equipment used in substation<br>2.5 Identifying control panels and protection relays<br>2.6 Recognizing efficiency losses in the voltage transformation process and specific voltage ratings for HT cables<br>2.7 Identifying HT cable types and LT cable types and their suitability for specific applications<br>2.8 Identifying earthing systems, components of earthing systems and atmospheric lightning protection systems (surge protection device) and lightning rod |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, subordinates and seniors in the workplace                                                                                                                                                                                                                                                                                                                                                                                           |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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|  | <p>4.2 Tools, equipment, materials and facilities appropriate to the process or activities</p> <p>4.3 Various kinds of electrical tools, equipment and instruments</p> <p>4.4 High Tension (HT) and Low tension (LT) cables, and protection relays</p> <p>4.5 Tools and equipment appropriate to joint and connection process</p> |
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### Assessment Evidence Guide

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| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <p>1.1 Described major equipment used in substation</p> <p>1.2 Explained atmospheric lightning protection systems, surge protection device (surge arrester)</p> <p>1.3 Identified transmission voltage levels, range of high-voltage transmission levels and range of medium-voltage transmission levels</p> <p>1.4 Identified major equipment used in substation</p> <p>1.5 Recognized efficiency losses in the voltage transformation process and specific voltage ratings for HT cables</p> <p>1.6 Identified earthing systems, components of earthing systems and atmospheric lightning protection systems (surge protection device) and lightning rod</p> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <p>2.1 Written test</p> <p>2.2 Practical demonstration</p> <p>2.3 Oral questioning</p> <p>2.4 Portfolio (Optional)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3. Context of Assessment          | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <b>Unit of Competency:</b><br><b>ASSEMBLE AND INSTALL DISTRIBUTION TRANSFORMERS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Nominal Duration:</b><br>80 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-03-O |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to assemble and install distribution transformers. It specifically includes the tasks of understanding electrical schematics and plan, gathering necessary tools and equipment, assembling the distribution transformer, installing the distribution transformer, connecting the transformer to electrical circuits, mounting and connecting switchgear to the transformer, and testing the installation and performing system checks. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                       | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Understand electrical schematics and plan | 1.1 Electrical schematics are reviewed and understood.<br>1.2 <b><u>Symbols</u></b> used in electrical schematics are recognized and understood.<br>1.3 Power distribution plans are understood.<br>1.4 Load calculations and current ratings indicated on electrical plans are understood.<br>1.5 Control circuit diagrams are reviewed and interpreted.                                                                                                                |
| 2. Gather necessary tools and equipment      | 2.1 Tools and equipment are identified and arranged based on the specific requirements of the job.<br>2.2 <b><u>Electrical testing devices</u></b> are collected.<br>2.3 <b><u>Measuring instruments</u></b> are gathered.<br>2.4 <b><u>Spare parts and materials</u></b> are identified.                                                                                                                                                                                |
| 3. Assemble the distribution transformer     | 3.1 <b><u>Transformer components</u></b> are inspected and prepared.<br>3.2 Core and windings are positioned for efficient operation.<br>3.3 Bushing fittings are installed.<br>3.4 Transformer tank and connected to the windings.<br>3.5 Tap changer components are assembled and tested.<br>3.6 Cooling system components are installed.<br>3.7 Transformer tank is sealed and pressure-tested.<br>3.8 Oil filling is done carefully, and the transformer is checked. |
| 4. Install the distribution transformer      | 4.1 Installation site for the distribution transformer is inspected and prepared.<br>4.2 Transformer foundation is checked for alignment and level.<br>4.3 Lifting equipment is used to safely move and position the transformer at the installation site.<br>4.4 Electrical connections are made between the transformer's bushings and the incoming/outgoing cables.<br>4.5 Grounding connections are installed and verified.                                          |



|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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|                                                    | <p>4.6 Oil filling is performed, and the transformer is checked for proper oil levels.</p> <p>4.7 <b>Control and protection devices</b> are installed and connected.</p> <p>4.8 Voltage testing is conducted on the transformer after installation.</p>                                                                                                                                                                                                                                                                                                              |
| 5. Connect the transformer to electrical circuits  | <p>5.1 Grounding connections are established and verified.</p> <p>5.2 Busbars or distribution panels are connected to the transformer's secondary terminals.</p> <p>5.3 <b>Switchgear</b> or circuit breakers are installed and connected to the transformer's primary and secondary sides.</p> <p>5.4 Voltage regulators or tap changers are connected and tested.</p> <p>5.5 Current transformers (CTs) and potential transformers (PTs) are connected for metering.</p>                                                                                           |
| 6. Mount and connect switchgear to the transformer | <p>6.1 Switchgear is selected according to the transformer's voltage and current ratings.</p> <p>6.2 Mounting area for the switchgear is inspected and prepared.</p> <p>6.3 Switchgear is properly mounted on the foundation or frame.</p> <p>6.4 Control wiring and auxiliary connections are made between the switchgear and the transformer.</p> <p>6.5 Grounding connections are established between the switchgear and the transformer.</p>                                                                                                                     |
| 7. Test the installation and perform system checks | <p>7.1 Installation is visually inspected to ensure all components are securely mounted and properly connected.</p> <p>7.2 Voltage levels are checked at various points in the system.</p> <p>7.3 Continuity tests are performed to ensure that all electrical connections.</p> <p>7.4 Grounding systems are checked for proper installation and resistance.</p> <p>7.5 Load testing is performed by applying the expected operational load to the transformer and switchgear.</p> <p>7.6 Control and monitoring systems are tested to ensure correct operation.</p> |

### Range of Variables

| Variable                      | Range (Includes but not limited to:)                                           |
|-------------------------------|--------------------------------------------------------------------------------|
| 1. Symbols                    | <p>1.1 Wires</p> <p>1.2 Switches</p> <p>1.3 Relays</p> <p>1.4 Transformers</p> |
| 2. Electrical testing devices | <p>2.1 Multimeters</p> <p>2.2 Circuit tester</p>                               |

|                                   |                                                                                                                                                         |
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|                                   | 2.3 Insulation testers<br>2.4 Megger<br>2.5 No-load and full load test device<br>2.6 Transformer oil test machine<br>2.7 Transformer ratio test machine |
| 3. Measuring instruments          | 3.1 Calipers<br>3.2 Voltmeters                                                                                                                          |
| 4. Spare parts and materials      | 4.1 Cables<br>4.2 Connectors<br>4.3 Fuses<br>4.4 Relay<br>4.5 CT<br>4.6 PT                                                                              |
| 5. Transformer components         | 5.1 Core<br>5.2 Windings<br>5.3 Bushings<br>5.4 Tap changer<br>5.5 Tank                                                                                 |
| 6. Control and protection devices | 6.1 Fuses<br>6.2 Circuit breakers<br>6.3 Relay                                                                                                          |
| 7. Switchgear                     | 7.1 MCCB<br>7.2 MCB<br>7.3 Busbar<br>7.4 Cable<br>7.5 Connector                                                                                         |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Electrical schematics, symbols, and power distribution plans<br>1.2 Load calculations & current ratings on electrical plans<br>1.3 Circuit diagrams                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2. Underpinning Skills    | 2.1 identifying and collecting tools and equipment, electrical testing devices and measuring instruments<br>2.2 Assembling distribution transformer including tap changer components, along with installing cooling system components<br>2.3 Sealing transformer tank, carefully performing oil filling and checking the transformer<br>2.4 Inspecting and preparing installation site for distribution transformer, checking transformer foundation, moving and positioning the transformer<br>2.5 Making electrical connections between transformer's bushings and incoming/outgoing cables, and installing & connecting control and protection devices<br>2.6 Connecting and testing voltage regulators or tap changers, |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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|                           | <p>and connecting current transformers (CTs) and potential transformers (PTs) for metering</p> <p>2.7 Properly mounting switchgear on the foundation or frame, and making control wiring and auxiliary connections between the switchgear and the transformer</p> <p>2.8 Establishing grounding connections between the switchgear and the transformer</p> <p>2.9 Performing continuity tests for all electrical connections, grounding systems for proper installation and resistance, load testing for the transformer and switchgear</p> |
| 3. Underpinning Attitudes | <p>1.1 Commitment to occupational health and safety</p> <p>1.2 Promptness in carrying out activities</p> <p>1.3 Sincerity and honesty to duties</p> <p>1.4 Environmental concerns</p> <p>1.5 Eagerness to learn</p> <p>1.6 Tidiness and timeliness</p> <p>1.7 Respect for rights of peers and seniors in workplace</p> <p>1.8 Good relationships with peers, sub-ordinates and seniors in workplace</p>                                                                                                                                     |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <p>1.1 Workplace (simulated or actual)</p> <p>1.2 Tools and equipment required for the job</p> <p>1.3 Electrical measuring instruments and testing devices</p> <p>1.4 Materials required</p> <p>1.5 Electrical schematics and diagrams</p> <p>1.6 Instructional manuals</p>                                                                                                                                                                                                                |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <p>1.1 interpreted electrical schematics, symbols and power distribution plans</p> <p>1.2 Interpreted control circuit diagrams</p> <p>1.3 Assembled distribution transformer including tap changer components, along with installing cooling system components</p> <p>1.4 Inspected and prepared installation site for distribution transformer, checked transformer foundation, moved and positioned the transformer</p> <p>1.5 Made electrical connections between transformer's bushings and incoming/outgoing cables, and installed &amp; connected control and protection devices</p> <p>1.6 Properly mounted switchgear on the foundation or frame, and made control wiring and auxiliary connections between the switchgear and the transformer</p> |
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|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2. Methods of Assessment | Competency should be assessed by:<br>2.1 Written Test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                      |
| 3. Context of Assessment | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

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| <b>Unit of Competency:</b><br><b>IDENTIFY EQUIPMENT MALFUNCTION<br/>IN SUBSTATION</b>                                                                                                                                                                                                                                                                                                                                                                  | <b>Nominal Duration:</b><br>80 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-04-O |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to identify equipment malfunction in substation. It specifically includes the tasks of performing visual inspection of substation components, checking transformer faults, checking low tension (LT) switchgear, checking high tension (HT) switchgear, checking capacitor bank (power factor improvement-PFI) and inspecting cable connections and earthing. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                                | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Perform visual inspection of substation components | 1.1 <b><u>Substation components</u></b> are identified for any signs of physical damage, wear, or corrosion.<br>1.2 Wiring and connections are checked.<br>1.3 Insulation materials are inspected for cracks, degradation, or physical damage.<br>1.4 Grounding systems are visually examined to ensure that all grounding connections are secure.<br>1.5 Cooling systems are visually inspected for proper function and any leaks. |
| 2. Check transformer faults                           | 2.1 Oil levels and quality are checked.<br>2.2 Transformer bushings are examined for cracks, wear, or discoloration.<br>2.3 Electrical connections at the primary and secondary terminals are checked.<br>2.4 Internal fault indicators are checked for activation or signs of faults.<br>2.5 Load current and voltage readings are taken.<br>2.6 Winding resistance and insulation resistance tests are performed.                 |
| 3. Check low tension (LT) switchgear                  | 3.1 Connection terminals are checked for tightness and corrosion.<br>3.2 Insulation integrity is examined to ensure no cracks and degradation.<br>3.3 Control and protection devices are inspected for proper alignment and functionality.<br>3.4 Tripping mechanisms of circuit breakers are tested.<br>3.5 Switchgear operations are tested to ensure they operate smoothly.<br>3.6 Metering and indicator systems are checked.   |

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|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Check high tension (HT) switchgear                  | 4.1 Insulation integrity is examined, ensuring that there are no cracks and degradation.<br>4.2 Electrical connections at the primary and secondary terminals are checked.<br>4.3 Circuit breakers and disconnect switches are inspected for proper operation.<br>4.4 Tripping tests are performed.<br>4.5 Operating mechanisms of the switchgear are tested to ensure smooth and reliable operation.<br>4.6 Earthing and grounding connections are checked.<br>4.7 Metering and indicator systems are tested.                                                |
| 5. Check capacitor bank (power factor improvement-PFI) | 5.1 Capacitor connections are inspected for tightness, corrosion, or loose connections.<br>5.2 Capacitance values are measured using appropriate testing equipment.<br>5.3 Voltage levels across the capacitor bank are checked.<br>5.4 Control relays are tested.<br>5.5 Harmonics and overvoltage conditions are monitored.<br>5.6 Fuses and circuit breakers are checked for proper functionality.<br>5.7 Cooling and ventilation systems are checked to ensure they are properly functioning.<br>5.8 Earthing system for the capacitor bank is inspected. |
| 6. Inspect cable connections and earthing              | 6.1 Cable connections are inspected for signs of wear, corrosion.<br>6.2 Cable insulation is checked for any visible damage, cuts, or abrasions.<br>6.3 <b>Connection points</b> are inspected.<br>6.4 Electrical conductors are checked for proper sizing and rated capacity.<br>6.5 Earthing connections are inspected.<br>6.6 Continuity tests are performed on cable connections and earthing systems.                                                                                                                                                    |

#### Range of Variables

| Variable                 | Range (Includes but not limited to:)                                      |
|--------------------------|---------------------------------------------------------------------------|
| 1. Substation components | 1.1 Transformers<br>1.2 Circuit breakers<br>1.3 Busbars<br>1.4 Switchgear |
| 2. Connection points     | 2.1 Terminal blocks<br>2.2 Busbars<br>2.3 Junction boxes                  |

## Curricular Content Guide

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|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <ul style="list-style-type: none"> <li>1.1 Substation components-transformers, circuit breakers, busbars and switchgear</li> <li>1.2 Low tension (LT) and high tension (HT) switchgears</li> <li>1.3 Capacitor bank (PFI-power factor improvement)</li> <li>1.4 Connection points-terminal blocks, busbars &amp; junction boxes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2. Underpinning Skills    | <ul style="list-style-type: none"> <li>2.1 Performing visual inspection of substation components-transformers, circuit breakers, busbars and switchgear for identifying any signs of physical damage, wear or corrosion</li> <li>2.2 Checking electrical wiring and connections at the primary and secondary terminals</li> <li>2.3 Examining transformer bushings for faults like cracks, wear or discoloration</li> <li>2.4 Testing operating mechanisms of the LT &amp; HT switchgears to ensure smooth and reliable operation</li> <li>2.5 Checking connection terminals for tightness and corrosion, and insulation integrity to ensure zero cracks and degradation (for LT &amp; HT switchgear)</li> <li>2.6 Inspecting control &amp; protection devices for proper alignment and functionality, and circuit breakers &amp; disconnect switches for proper operation</li> <li>2.7 Inspecting/checking capacitor connections for tightness, corrosion or loose connections, voltage levels across the capacitor bank, and earthing system for the capacitor bank</li> <li>2.8 Inspecting connection points (terminal blocks, busbars &amp; junction boxes), and electrical conductors for proper sizing and rated capacity</li> </ul> |
| 3. Underpinning Attitudes | <ul style="list-style-type: none"> <li>3.1 Commitment to occupational health and safety</li> <li>3.2 Promptness in carrying out activities</li> <li>3.3 Sincerity and honesty to duties</li> <li>3.4 Environmental concerns</li> <li>3.5 Eagerness to learn</li> <li>3.6 Tidiness and timeliness</li> <li>3.7 Respect for rights of peers and seniors in workplace</li> <li>3.8 Good relationships with peers, sub ordinates and seniors in workplace</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <ul style="list-style-type: none"> <li>4.1 Workplace (simulated or actual)</li> <li>4.2 Tools, equipment and facilities needed for the process/activity</li> <li>4.3 Electrical measuring instruments</li> <li>4.4 Materials required</li> <li>4.5 Instructional manual</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Assessment Evidence Guide

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|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Described substation components (transformers, circuit breakers, busbars and switchgears)<br>1.2 Explained connection points (terminal blocks, busbars & junction boxes)<br>1.3 Performed visual inspection of substation components for identifying any signs of physical damage, wear or corrosion<br>1.4 Tested operating mechanisms of the LT & HT switchgears to ensure smooth and reliable operation<br>1.5 Checked connection terminals for tightness and corrosion, and insulation integrity to ensure zero cracks and degradation (for LT & HT switchgear)<br>1.6 Inspected capacitor connections for tightness, corrosion or loose connections, voltage levels across the capacitor bank, and earthing system for the capacitor bank |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



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| <b>Unit of Competency:</b><br><b>TROUBLESHOOT MALFUNCTION IN SUBSTATION</b>                                                                                                                                                                                                                                                                                                                                                       | <b>Nominal Duration:</b><br>60 hrs. | <b>Unit Code:</b><br>SICIP-LE-SIM-05-O |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to troubleshoot malfunction in substation. It specifically includes the tasks of performing initial diagnostic checks, testing and servicing substation components, checking and servicing control and protection systems, performing continuity testing on electrical circuits, and testing transformer and switchgear for performance. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                              | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Perform initial diagnostic checks                | 1.1 <b><u>Diagnostic tools</u></b> are identified and used.<br>1.2 Power supply to the system is verified to ensure it is functioning correctly.<br>1.3 Fuses, relays, and circuit breakers are inspected for signs of damage or failure.<br>1.4 Wiring and connections are visually inspected or disconnected wires.<br>1.5 <b><u>System readings</u></b> are taken.<br>1.6 Error codes or fault indications are recovered from the system's control module.<br>1.7 Mechanical systems are manually tested.                                                                                          |
| 2. Test and service substation components           | 2.1 <b><u>Substation equipment components</u></b> are identified.<br>2.2 <b><u>Transformer testing</u></b> is performed.<br>2.3 Circuit breakers are tested and serviced for proper tripping functionality.<br>2.4 Protective relays are tested and serviced to verify that they correctly detect fault conditions.<br>2.5 Busbars are tested and serviced for continuity and proper conductivity.<br>2.6 Control and monitoring systems are tested and serviced to verify that the equipment is correctly connected.<br>2.7 Grounding systems are tested and serviced for continuity and resistance. |
| 3. Check and service control and protection systems | 3.1 Control panels protection relays are checked and serviced.<br>3.2 Fuses and circuit breakers are checked and serviced.<br>3.3 Alarm and monitoring systems are tested and serviced.<br>3.4 Testing of protection devices is performed.<br>3.5 Protection devices are serviced.                                                                                                                                                                                                                                                                                                                    |

|                                                      |     |                                                                                                            |
|------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------|
|                                                      | 3.6 | Grounding and earthing connections are inspected and serviced.                                             |
| 4. Perform continuity testing on electrical circuits | 4.1 | Testing equipment is prepared and set to the appropriate settings for continuity testing.                  |
|                                                      | 4.2 | Power supply to the circuit is turned off.                                                                 |
|                                                      | 4.3 | Test leads are connected to the multimeter or continuity tester.                                           |
|                                                      | 4.4 | Circuit paths are checked for continuity.                                                                  |
|                                                      | 4.5 | Short circuits are checked.                                                                                |
|                                                      | 4.6 | Insulation resistance is tested.                                                                           |
|                                                      | 4.7 | After testing, the power supply is restored, and the electrical circuit is rechecked for normal operation. |
| 5. Test transformer and switchgear for performance   | 5.1 | Insulation resistance is measured using a megohmmeter.                                                     |
|                                                      | 5.2 | Load testing is performed on the transformer.                                                              |
|                                                      | 5.3 | Oil temperature is monitored.                                                                              |
|                                                      | 5.4 | Tap changer operation is tested.                                                                           |
|                                                      | 5.5 | Transformer oil quality is checked to avoid moisture content and contamination.                            |
|                                                      | 5.6 | Protection relays in the switchgear are checked and serviced.                                              |
|                                                      | 5.7 | Switchgear operation is tested.                                                                            |

### Range of Variables

| Variable                           | Range (Includes but not limited to:)                                                                                        |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1. Diagnostic tools                | 1.1 Multimeters<br>1.2 Oscilloscopes<br>1.3 Diagnostic scanners                                                             |
| 2. System readings                 | 2.1 Voltage<br>2.2 Current<br>2.3 Resistance                                                                                |
| 3. Substation equipment components | 3.1 Transformers<br>3.2 Circuit breakers<br>3.3 Relays<br>3.4 Busbars                                                       |
| 4. Transformer testing             | 4.1 Insulation resistance<br>4.2 Voltage ratio<br>4.3 No load test<br>4.4 Full load test<br>4.5 Vector test<br>4.6 Oil test |

### Curricular Content Guide

|                           |                                                                           |
|---------------------------|---------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Diagnostic tools – multimeters, oscilloscopes and diagnostic scanners |
|---------------------------|---------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 1.2 System readings – voltage, current & resistance<br>1.3 Substation equipment components – transformers, circuit breakers, relays and busbars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2. Underpinning Skills    | 2.1 Identifying and using diagnostic tools (multimeters, oscilloscopes and diagnostic scanners)<br>2.2 Inspecting power supply to the system to ensure its correct functioning, and fuses, relays and circuit breakers for signs of damage or failure<br>2.3 Identifying substation equipment components<br>2.4 Performing Transformer testing<br>2.5 Checking and servicing control systems<br>2.6 Checking and servicing protection systems<br>2.7 Preparing and setting testing equipment appropriately and performing continuity testing on electrical circuits<br>2.8 Testing transformer operation, and checking transformer oil quality to avoid moisture content and contamination<br>2.9 Testing switchgear operation, and checking and servicing protection relays in the switchgear |
| 3. Underpinning Attitudes | 2.1 Commitment to occupational health and safety<br>2.2 Promptness in carrying out activities<br>2.3 Sincerity and honesty to duties<br>2.4 Environmental concerns<br>2.5 Eagerness to learn<br>2.6 Tidiness and timeliness<br>2.7 Respect for rights of peers and seniors in workplace<br>2.8 Good relationships with peers, sub-ordinates and seniors in workplace                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Tools, equipment and facilities needed for the process/activity<br>4.3 Electrical measuring instruments<br>4.4 Materials relevant to the proposed activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Identified and used diagnostic tools<br>1.2 Identified substation equipment components<br>1.3 Checked and serviced control and protection systems<br>1.4 Prepared and set testing equipment appropriately and performed continuity testing on electrical circuits |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                          |                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>1.5 Tested transformer operation, and checked transformer oil quality to avoid moisture content and contamination</p> <p>1.6 Tested switchgear operation, and checked and serviced protection relays in the switchgear</p>                                            |
| 2. Methods of Assessment | <p>Competency should be assessed by:</p> <p>2.1 Written test</p> <p>2.2 Practical demonstration</p> <p>2.3 Oral questioning</p> <p>2.4 Portfolio (Optional)</p>                                                                                                          |
| 3. Context of Assessment | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p> |

### **Workshop/Lab Facility Standard**

|                            |                                      |
|----------------------------|--------------------------------------|
| <b>Course Name:</b>        | Substation Maintenance and Servicing |
| <b>Number of Trainees:</b> | 25                                   |

#### **Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):**

- Classroom – 350 sft
- Workshop/ lab – 1000 sft

#### **Major Training Equipment and Training Facilities:**

| <b>Sl. No.</b> | <b>Major Equipment and Training facilities</b>                                 | <b>Required facilities</b> |
|----------------|--------------------------------------------------------------------------------|----------------------------|
| 1.             | Computer/Laptop with internet facility                                         | 01                         |
| 2.             | Multimedia projector & screen                                                  | 01                         |
| 3.             | Transformer (100KVA)-11kV/0.4kV                                                | 01                         |
| 4.             | Transformer coil winding machine                                               | 02                         |
| 5.             | Transformer core cutting machine                                               | 02                         |
| 6.             | Main Distribution Board with busbar, CB and other accessories (Training model) | 02                         |
| 7.             | Sub Distribution Board with busbar, CB and other accessories (Training model)  | 02                         |
| 8.             | HT switchgear panel                                                            | 01                         |
| 9.             | LT switchgear panel                                                            | 01                         |
| 10.            | PFI panel                                                                      | 01                         |
| 11.            | Magnetic contactor (coil voltage 220V and 400V)                                | 10                         |
| 12.            | Relay                                                                          | 05                         |
| 13.            | Timer                                                                          | 05                         |
| 14.            | Pilot lamp                                                                     | 06                         |
| 15.            | Push switch                                                                    | 06                         |
| 16.            | IDMT relay                                                                     | 02                         |
| 17.            | Omron relay                                                                    | 02                         |
| 18.            | Phase sequence relay                                                           | 02                         |
| 19.            | Electric drill machine                                                         | 02                         |
| 20.            | Electric grinding machine                                                      | 02                         |
| 21.            | Soldering Stations                                                             | 02                         |
| 22.            | CT                                                                             | 03                         |
| 23.            | PT                                                                             | 02                         |
| 24.            | Indicating lamp                                                                | 03                         |

|     |                        |    |
|-----|------------------------|----|
| 25. | Multi meter/AVO meter  | 10 |
| 26. | Earth tester           | 02 |
| 27. | Digital weight machine | 01 |
| 28. | Insulation tester      | 02 |
| 29. | Tachometer             | 02 |
| 30. | Clamp Meter            | 02 |
| 31. | Panel lock             | 02 |
| 32. | Air vent               | 02 |
| 33. | Exhaust fan            | 02 |
| 34. | Insulator              | 08 |
| 35. | Copper (12')           | 01 |
| 36. | Copper tube (15')      | 03 |

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Substation Maintenance & Servicing, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.